Contact

www.linkedin.com/in/ryan-burch-ai (LinkedIn)

Top Skills

Software Design
Large Scale Development
Artificial Intelligence (AI)

Ryan Burch

Staff AI/ML | Forward Deployed Engineer | Generative AI & LLM Systems Architect | RAG, Multi-Agent AI, LangChain/LangGraph | AWS Bedrock, Kinesis, Lambda | Scalable AI Systems
United States

Summary

I am a Senior/Staff-level AI/ML Engineer and Generative AI Architect with a strong focus on building production-grade, enterprise-scale intelligent systems powered by Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and cloud-native architectures.My expertise lies in designing end-to-end AI platforms that seamlessly integrate data pipelines, vector search, and advanced reasoning systems, enabling organizations to unlock actionable insights and automate complex workflows. Core ExpertiseArchitecting LLM-powered applications with advanced prompt engineering, contextual memory, and reasoning workflowsDesigning RAG pipelines with optimized retrieval strategies, hybrid search, and embedding lifecycle managementBuilding multi-agent AI systems using LangGraph, enabling autonomous decision-making and orchestrationDeveloping real-time AI platforms using event-driven architectures (Kinesis, Kafka, Flink)Implementing AI governance frameworks including observability, auditability, traceability, and compliance (GDPR, SOX)Optimizing model performance, latency, and cost efficiency in production environments. Cloud & AI InfrastructureI specialize in deploying AI systems on AWS, leveraging:Amazon Bedrock (foundation models, orchestration, guardrails)Serverless architectures (Lambda, API Gateway)Data pipelines (Kinesis, S3, Glue)Storage systems (DynamoDB, Aurora, OpenSearch)I design systems that are:Highly scalable and fault-tolerantEvent-driven and loosely coupledSecure and compliance-ready Advanced AI EngineeringVector databases & semantic retrieval (FAISS, Pinecone, OpenSearch, Redis)Embedding optimization & re-ranking strategiesFine-tuning strategies and model adaptationAI evaluation pipelines (LangSmith, custom metrics, human-in-the-loop)Guardrails, hallucination mitigation, and safe AI deployment Software & System DesignDistributed systems & microservices architectureHigh-throughput, low-latency APIsReal-

time stream processing & data engineeringScalable backend systems using FastAPI, Django, Node.jsInfrastructure as Code (Terraform, Docker, Kubernetes)

Experience

Johnson & Johnson
Staff AI/ML Engineer/FDE
January 2023 - Present (3 years 10 months)

Stripe
Staff AI Engineer
July 2018 - January 2023 (4 years 7 months)

Ellipsis Health
Machine Learning Specialist
April 2014 - June 2018 (4 years 3 months)
New York, New York, United States

- Developed and maintained applications supporting sustainability and compliance initiatives
- Worked closely with product owners and business stakeholders
- Helped ensure data consistency and integrity across platform workflows
- Improved the reliability and maintainability of existing systems
- Contributed to planning and delivery of new features
- Assisted with monitoring, issue investigation, and system support
- Took part in continuous improvement of development practices

Education

The King's College
Master's Degree, Computer Science

The University of Texas at Arlington
Bachelor of Science